

Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$	1			1		
2.				$[\text{Cr}(\text{OH})_6]^{3-}$	1			1		
3.	(a)			$6\text{NaOH} + 3\text{Cl}_2 \rightarrow \text{NaClO}_3 + 5\text{NaCl} + 3\text{H}_2\text{O}$	1			1		
	(b)			disproportionation where the same element is both oxidised and reduced	1			1		
4.				(white precipitate and) steamy fumes	1			1		1
5.				0.0093 mol dm ⁻³ (1) 2.36 g dm ⁻³ (1)		2		2	1	
6.				(Gibbs free energy is negative) as entropy increase of solution overcomes endothermic enthalpy change		1		1		
7.				pH between 2 and 6.5 (1) ammonium ion dissociates to release H ⁺ ions (1)		2		2		
				Section A total	5	5	0	10	1	1